Day 38 EBIKE101 Instructor Run Sheet

Program: CTS Cohort 6

Course: EBIKE101

Week: Final week of CTS / Week 2 of EBIKE101

Date: Tuesday, 2026-09-08

Context: Monday 2026-09-07 was a holiday. Resume with a practical diagnostic day.

Daily focus: Diagnose the E-bike Homunculous, identify what is needed to finish it, and begin safe finish work.

Big idea

A project is not “almost done” because it looks like a bike. It is close to done when the team can explain what is installed, what is missing, what is verified, what is unsafe, and what the next safe work step is.

Student-facing goal

By the end of today, students should be able to:

1. Inventory an unfinished e-bike build.
2. Separate mechanical problems from electrical/control problems.
3. Identify missing parts or unknowns.
4. Trace source, protection, control, and load paths.
5. Create a finish-work list that is safe, ordered, and realistic.
6. Begin work only on tasks that are safe and approved.

Materials

- E-bike Homunculous project
- Basic hand tools
- PPE
- Multimeters
- Zip ties / wire management materials
- Labels / tape / marker
- Notebook or printed diagnostic worksheet
- Camera or phone for evidence photos
- Known-good charger/battery info if available
- Any available controller/display/PAS/throttle documentation

Safety boundary

Students may inspect, document, trace, label, and perform approved mechanical/wire-management work. Battery connection, powered testing, controller testing, and motor testing require instructor control.

Do not allow:

- Battery shorting experiments

- Continuity testing on live battery circuits
- Bypassing fuses
- Full-throttle tests
- Ride testing before mechanical and electrical checks
- Hands near chain, spokes, wheel, or drivetrain during powered testing

Schedule

10:00–10:15 — Arrival / reset / context

- Remind students Monday was a holiday and today restarts the build thread.
- Name the project: E-bike Homunculous diagnosis and finish work.
- Frame the day as “find out what this bike needs,” not “make it work at any cost.”

Instructor line:

“We are not trying to prove the bike works. We are trying to prove what is true about it.”

10:15–10:35 — Morning quiz

- Give student quiz.
- Collect or review immediately.
- Use missed questions to decide reteach focus.

10:35–10:55 — Safety and system map mini-lesson

Cover:

- Source: battery/charger
- Protection: fuse/breaker/BMS/controller limits
- Control: display, PAS, throttle, brake cutoff, key/switch
- Load: motor, lights, accessories
- Frame/mechanical: brakes, wheels, torque arms, chain, mounts

Key point:

A silent motor does not prove a bad motor. It can be missing power, missing control input, active brake cutoff, bad connector, wrong display/controller pairing, bad PAS, blown fuse, or mechanical lockup.

10:55–11:30 — Whole-bike inventory

Students work in pairs or one full group depending on class size.

Inventory categories:

- Frame and mounting
- Wheels/tires/brakes
- Battery and mount
- Controller and mount
- Display/control parts
- PAS/throttle/brake cutoffs
- Motor wiring
- Connectors
- Protection/fusing
- Missing parts
- Unknowns

11:30–12:00 — Diagnostic board

Create a board with four columns:

1. Verified
2. Unknown
3. Missing/Needed
4. Unsafe until fixed

Students must support every claim with observation, measurement, or documented uncertainty.

12:00–12:30 — Lunch / reset

Before break, make the bike safe:

- Battery disconnected or controlled by instructor
- Tools reset
- Loose parts contained
- Wires not left under tension

12:30–1:00 — Finish-work plan

As a class, choose approved tasks:

- Mechanical tightening/inspection
- Brake adjustment
- Cable routing
- Labeling
- Connector inspection
- Mounting cleanup
- Parts list
- Documentation/photos
- Non-powered continuity checks only where appropriate

Do not jump straight to powered testing because impatience is not an electrical method.

1:00–2:20 — Approved work block

Students rotate through roles:

- Lead mechanic: physical fitment, fasteners, brakes
- Electrical tracer: source/control/load path map
- Scribe: worksheet, parts list, evidence log
- Safety observer: pinch points, sharp edges, battery boundary
- Parts runner: organizes missing/needed components

Instructor circulates and signs off before any higher-risk work.

2:20–2:45 — Controlled test decision

Do not guarantee powered testing. Decide from evidence.

Powered test may happen only if:

- Wheel is secure
- Brakes function

- Axle/torque reaction is controlled
- Battery voltage/rating is appropriate
- Fuse/protection path is present
- No damaged/melted wiring
- Connectors are seated and strain-relieved
- Drive wheel is safely lifted or restrained
- Instructor controls first power-up

2:45–3:00 — Closeout

Students complete:

- What we verified
- What is still unknown
- Parts needed
- Next safe action
- Photo evidence list

Instructor closeout notes

Today is successful if the class leaves with a clear diagnostic state, not necessarily a finished bike. That distinction will save you from the ancient human urge to call chaos “progress” because something briefly spun.